Research Portfolio

HANG ZHANG

ETH Zurich
Digital Building Technologies



CONTENTS

RESEARCH

02 MaskPLAN
CVPR 2024

04 MaskLayout
Automation in Construction (2026)

05 TextPLAN
Automation in Construction (2026)

07 FreeLayout
Advanced Engineering Informatics (2026)

09 FabPLAN
Under review

11 Text-to-Form
ACADIA 2020

12 3D Model Generation
Technologies 2019

13 3D Form Style Transfer
CAADRIA 2020

14 Composite Pillar
Extreme Mechanics Letters 2022

15 Directional Adhesion
Under review

16 2D–3D Form Finding
DigitalFUTURES CDRF 2020 · Springer 2021

18 Wave Function Collapse
CAADRIA PGSC 2020 · presentation

DESIGNS

21 Future Airport

22 Light Tracing

23 Schizobulia House

24 Diffusion Structure

25 Fashion in Arch

Select a project title to jump to its opening slide.

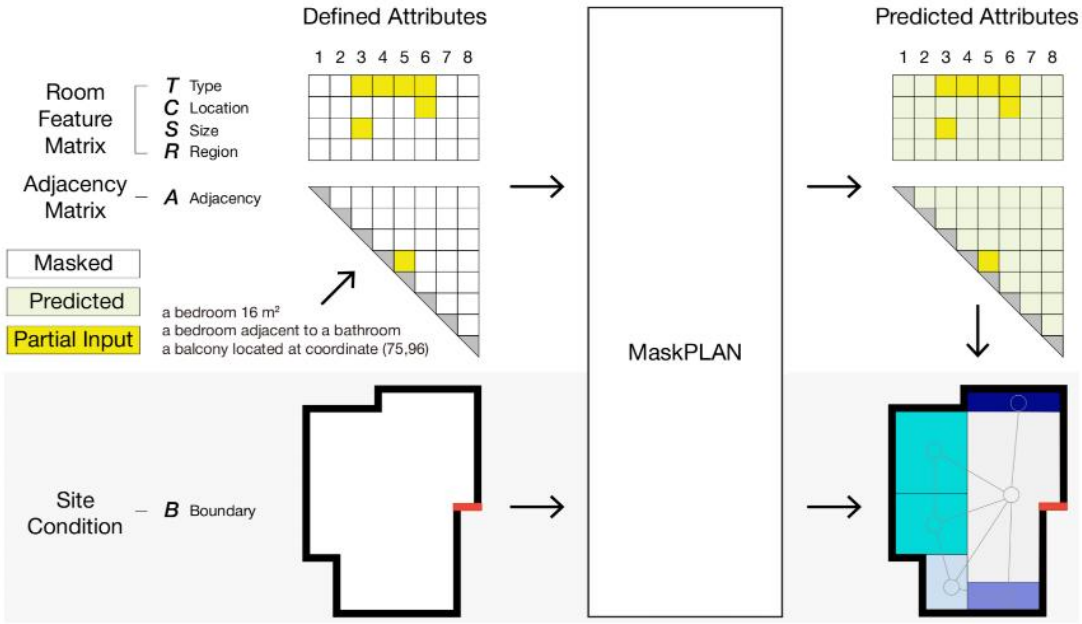
MaskPLAN

Masked generative layout planning from partial input

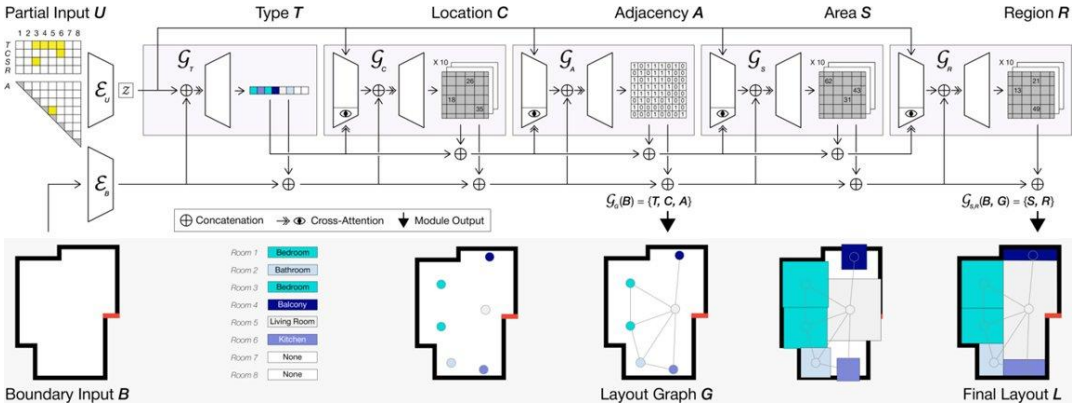
CVPR 2024 · FIRST AUTHOR

From incomplete intent to a complete, editable plan.

MaskPLAN reframes floorplan generation as **completion** rather than one-shot synthesis. A **graph-structured dynamic masked autoencoder** accepts any subset of **five cross-dependent attributes**—room type, location, adjacency, area, and region—then predicts what is missing while respecting the site boundary.



Layout planning from free-range partial user input

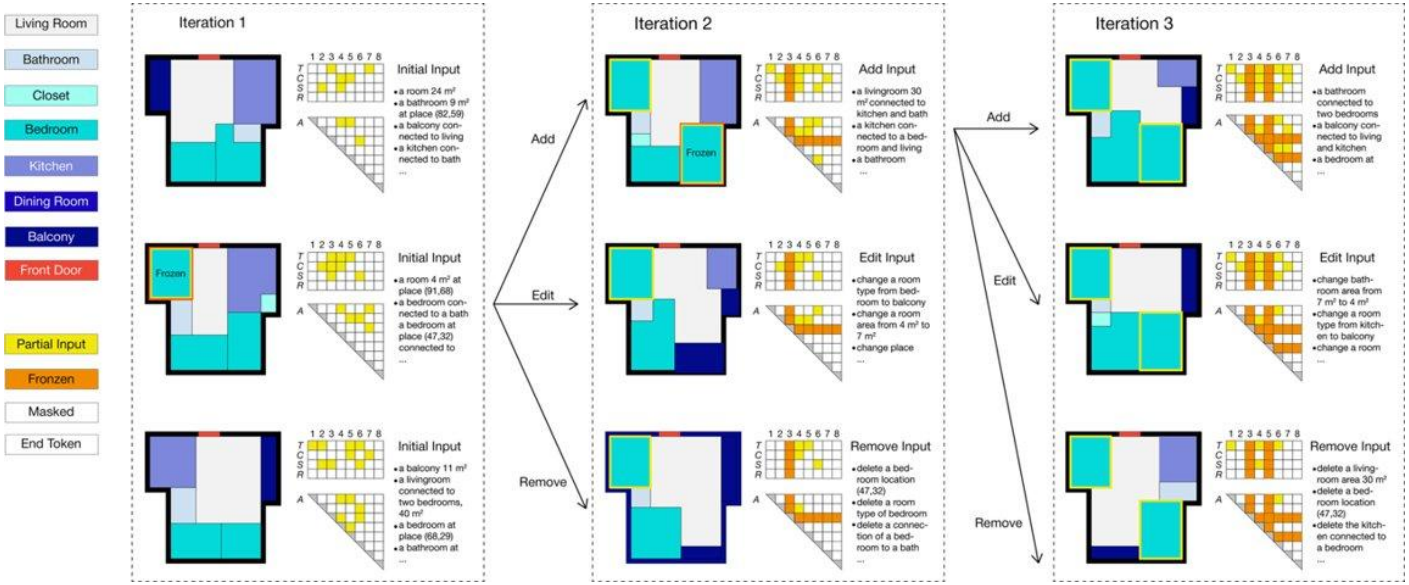


Framework: the partial input encodes a global prior across five attribute-specific generators.

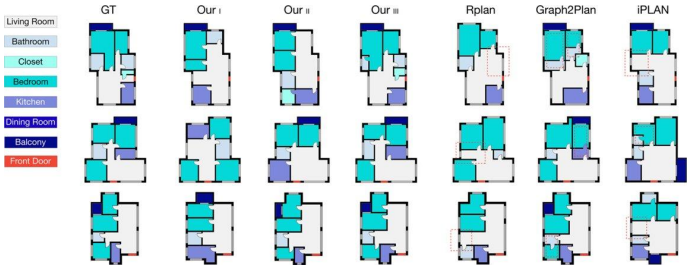
READ FULL PAPER ↗

MaskPLAN

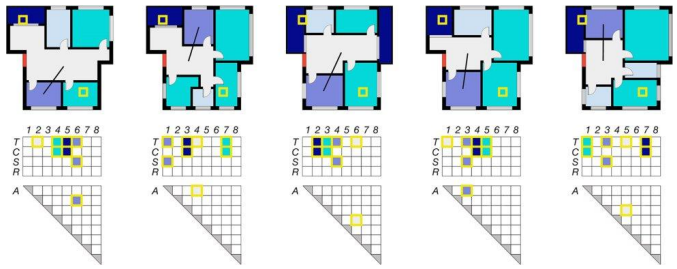
Iteration is the interface: add, edit, remove, freeze, branch



Iterative customization: add, edit, remove, or freeze attributes to explore design paths.



Reconstruction vs. baselines; missing or blocked rooms marked in red.



Alternatives from one partial input by varying attribute order.

Generative assistance stays reversible.

Users can progressively **add**, **edit**, **remove**, or **freeze** design decisions. **Alternative generations** preserve the same partial brief while varying unresolved attributes. This turns generation into a conversational design loop rather than a terminal output.

READ FULL PAPER ↗

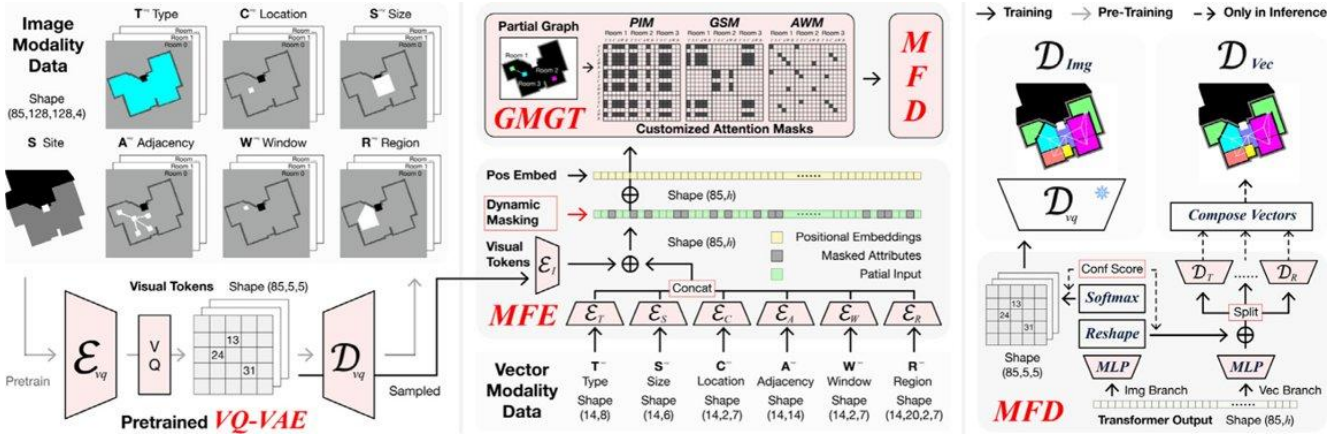
MaskLayout

Dual-modality masked generation for irregular apartment layouts

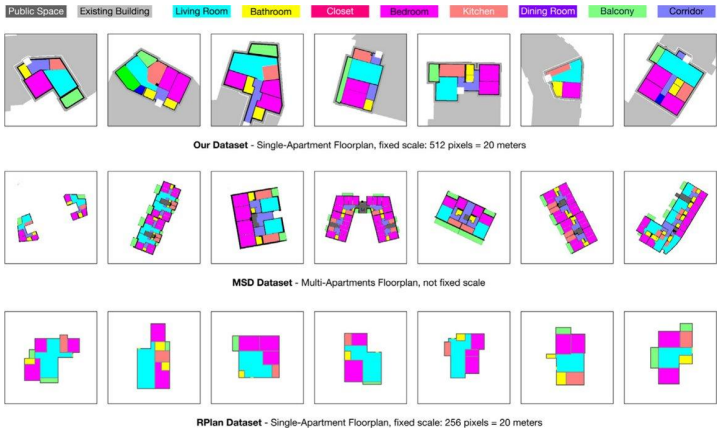
AUTOMATION IN CONSTRUCTION 190 (2026) 107120

Six layout attributes, learned in vector and image space.

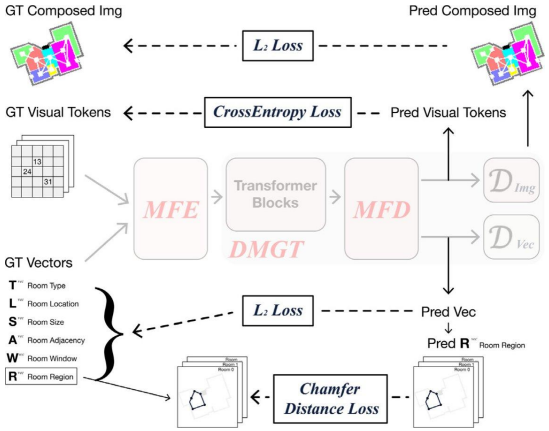
MaskLayout pairs a **43,000+ apartment benchmark** with a bidirectional transformer that fuses **vector and image space**. The hybrid representation covers **six attributes**—room type, location, size, adjacency, window, and region—while **dynamic masking** enables refinement from sparse partial input on irregular boundaries.



Framework: a bidirectional transformer with modality-fusion encoders and decoders.



A single-apartment benchmark compared with MSD and RPLAN.



Hybrid training loss across image and vector modalities.

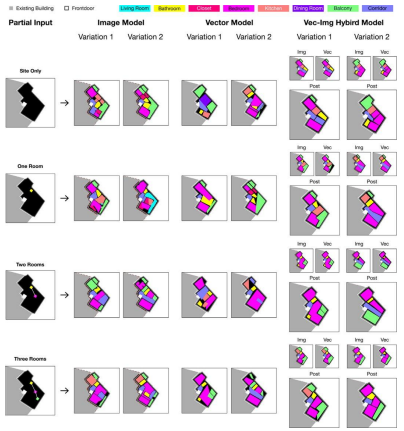


Image-only, vector-only, and hybrid dual-modality variants.

READ FULL PAPER ↗

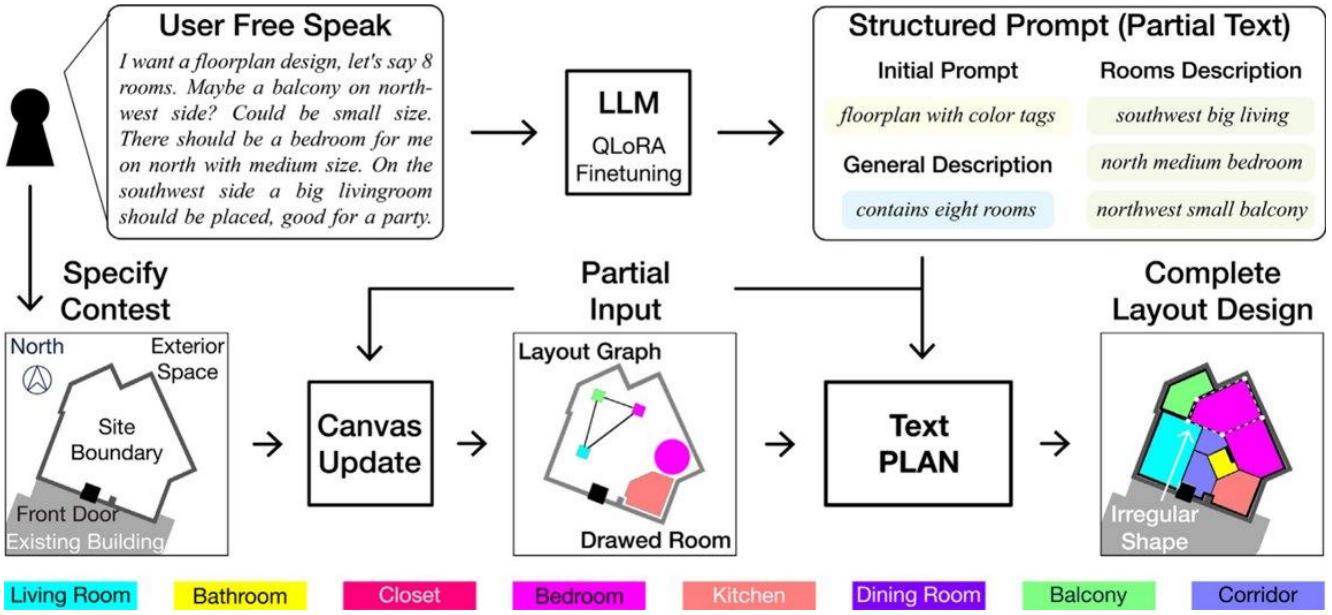
TextPLAN

Multimodal floorplan synthesis through language, graphs, and sketches

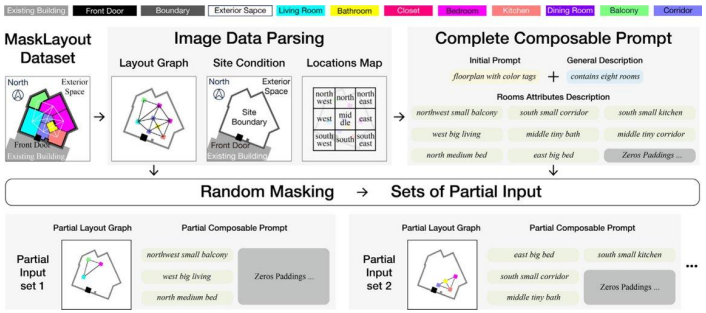
AUTOMATION IN CONSTRUCTION 190 (2026) 107037

Architectural intent can enter as speech, structure, or drawing.

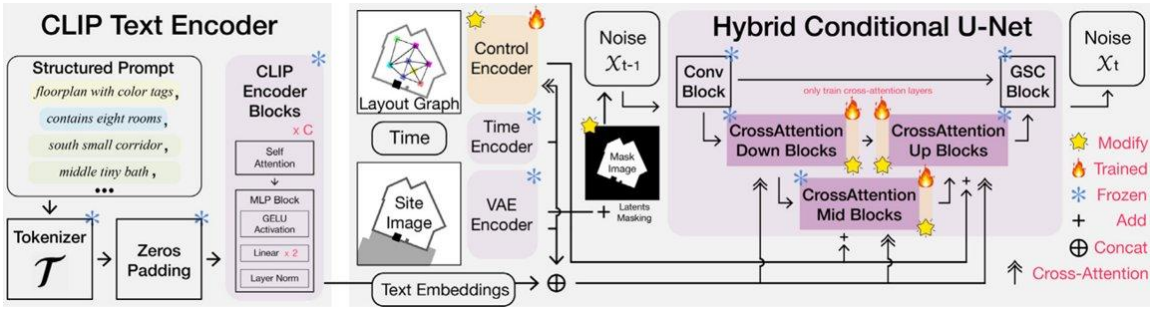
TextPLAN converts free-form instructions into structured prompts using a **QLoRA-adapted language model**, then combines them with **partial graphs**, **site boundaries**, and **user sketches**. An **SDXL backbone** with **composable diffusion** preserves independently weighted constraints across modalities.



Overview: speech, structure, and sketches converge into one layout pipeline.



Data preprocessing pipeline.

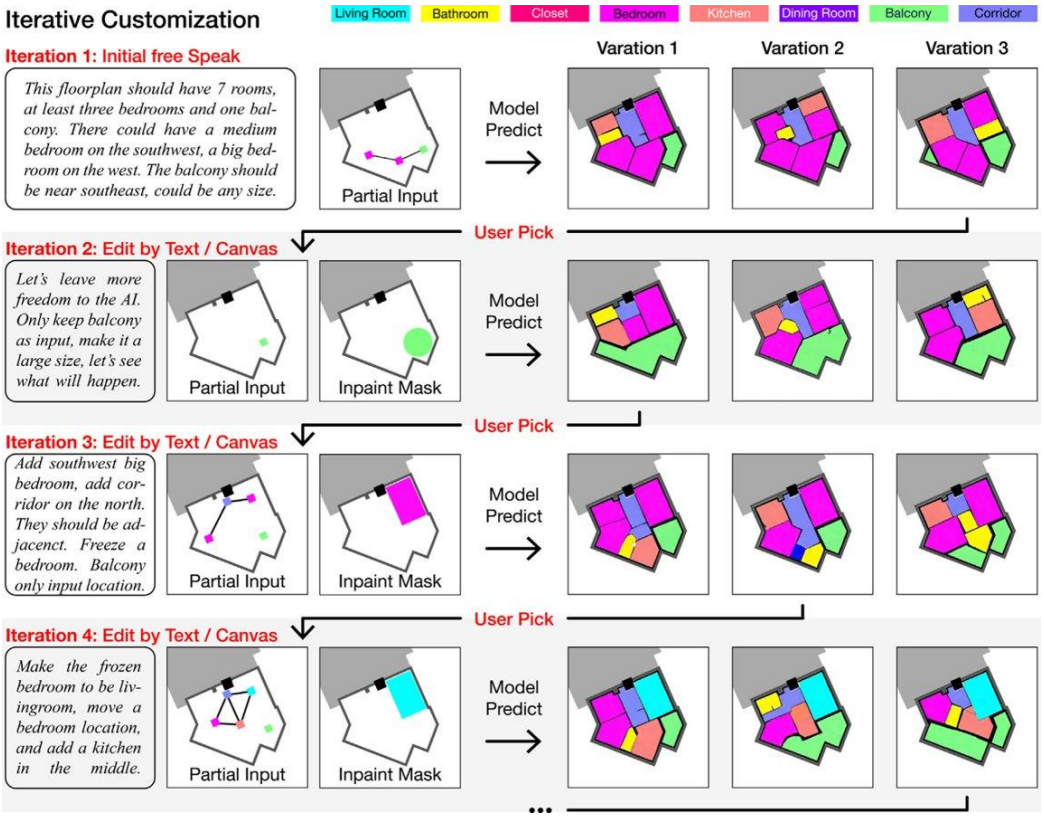


The TextPLAN architecture: a hybrid conditional U-Net.

READ FULL PAPER ↗

TextPLAN

A multimodal design loop with measurable reconstruction evidence

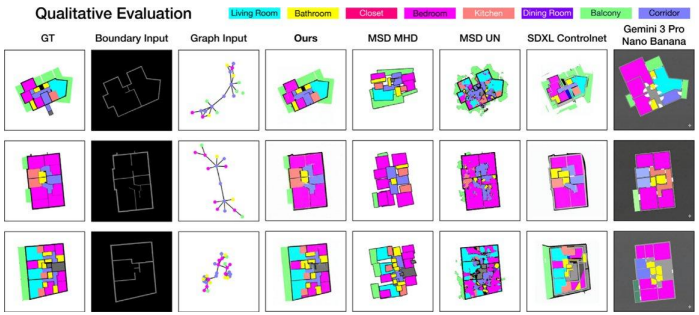


Iterative customization preserves prior intent across language, graph, and sketch edits.

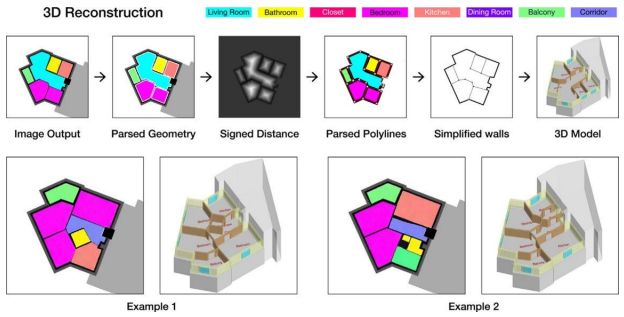
Generation remains controllable after the first result.

TextPLAN supports **language-led revisions**, **sketch edits**, and **partial-layout continuation** within one workflow. Comparative tests show stronger response to irregular boundaries and multimodal constraints; the same output is then **vectorized** and reconstructed as an architectural **3D model**.

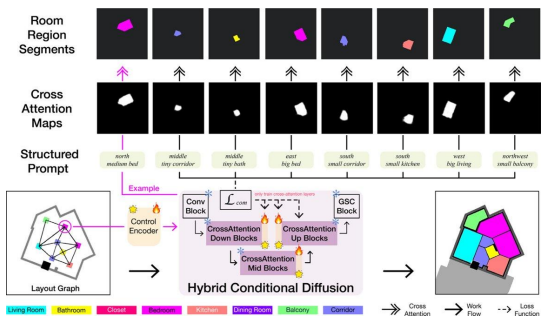
READ FULL PAPER ↗



Qualitative comparison with baselines under irregular sites.



Vectorization closes the loop from raster output to a 3D model.



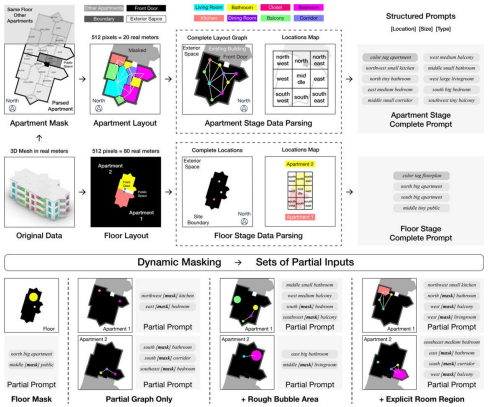
Composable diffusion weights constraints independently.

FreeLayout

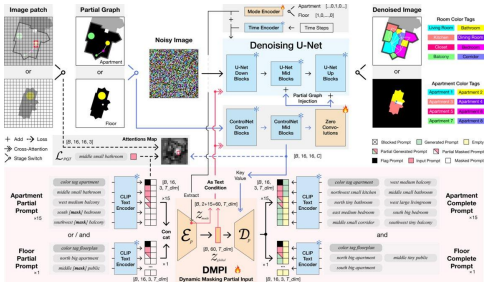
Free-range multimodal input for free-form building layout generation

One diffusion system, two hierarchical design scales.

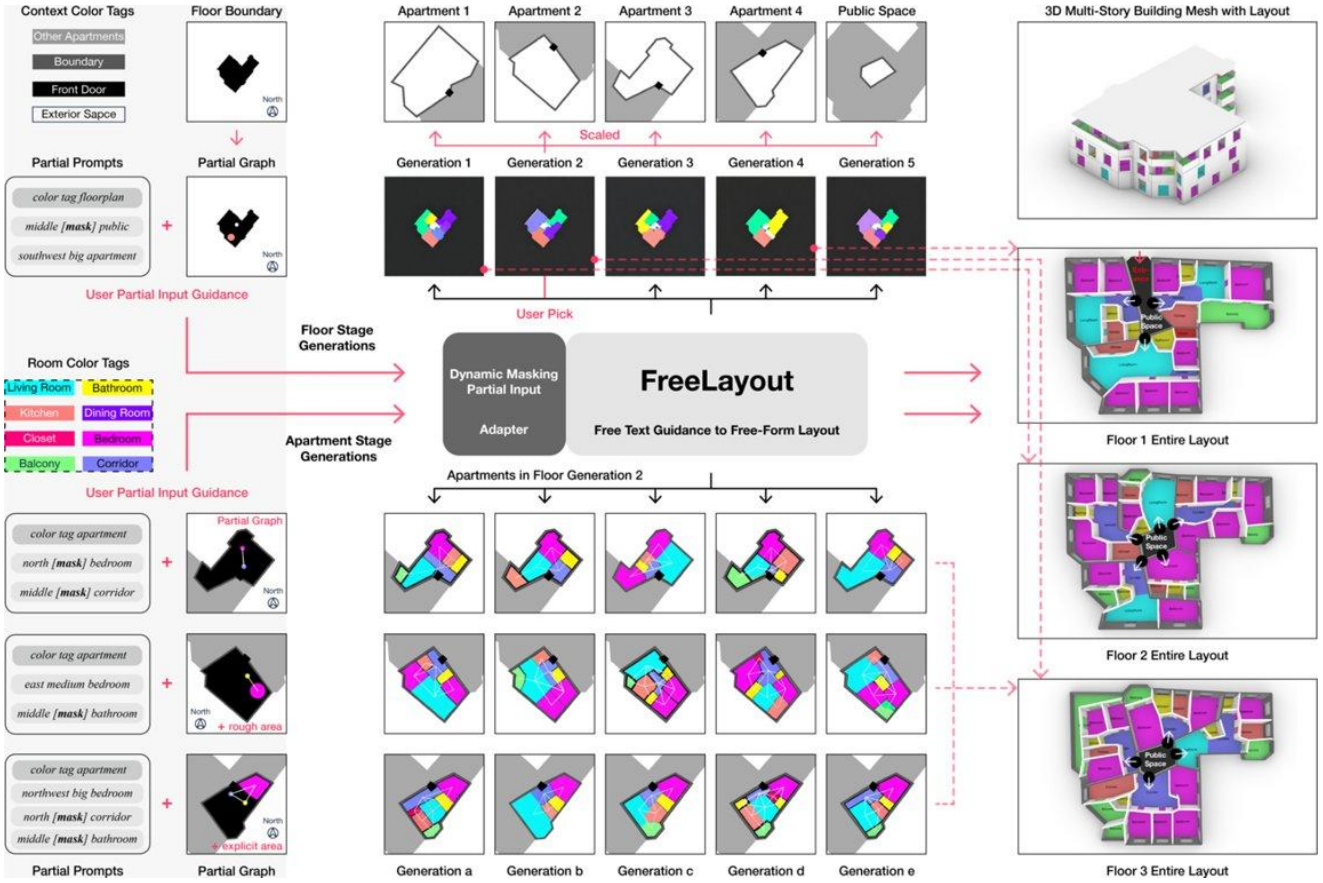
FreeLayout extends partial-input generation from apartments to **whole floors**. A **shared SDXL model** switches between floor segmentation and room-layout modes while **Dynamic Masking Partial Input** aligns structured prompts, partial graphs, rough bubbles, explicit regions, and site context.



Aligned multimodal data: prompts, graphs, sketches, and masks.



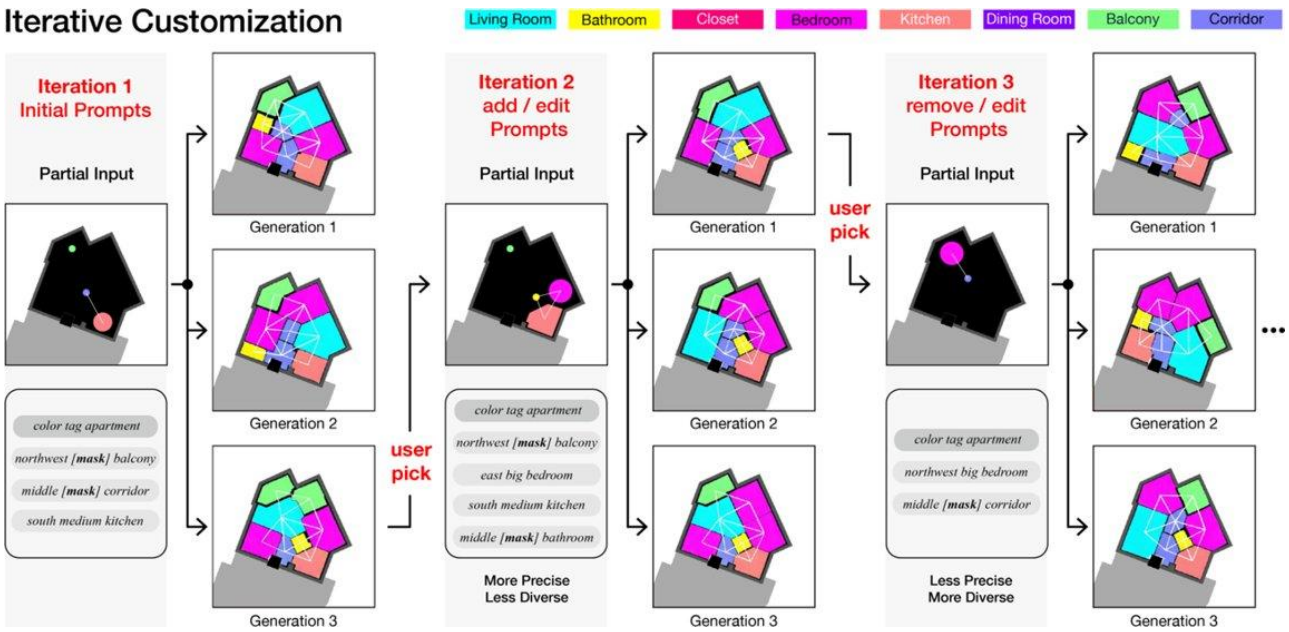
A shared SDXL backbone with a mode switcher and DMPI.



FreeLayout: free-form building-layout planning under partial multimodal guidance.

FreeLayout

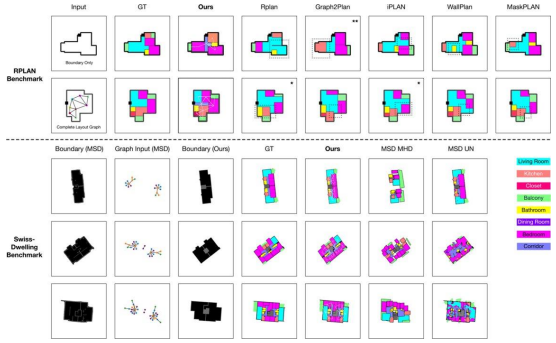
Hierarchical generation that remains open to correction



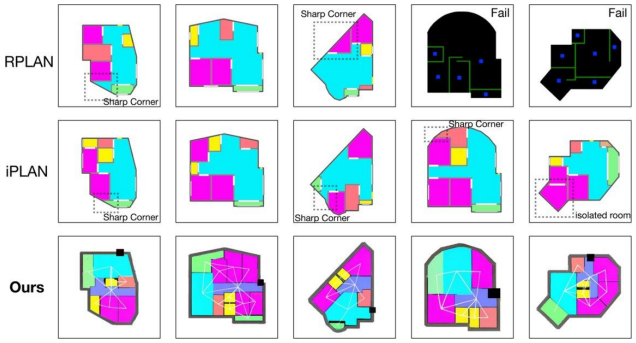
Iterative customization: add, edit, or remove constraints across modalities.

Partial inputs remain synchronized across hierarchy.

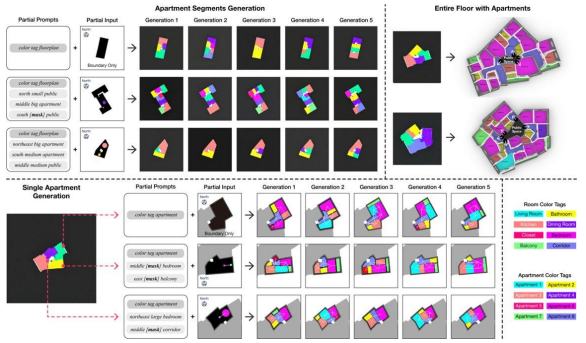
Dynamic masking accepts **boundary-only** briefs or richer combinations of **text, graphs, sketches, and room regions**. Comparisons show robustness on **irregular footprints**; simplified **3D reconstruction** demonstrates a path from generated hierarchy to building form.



Qualitative comparisons on RPLAN and SwissDwellings.



Boundary-aware room orientation avoids sharp residual corners.



Additional floor- and apartment-level generation results.

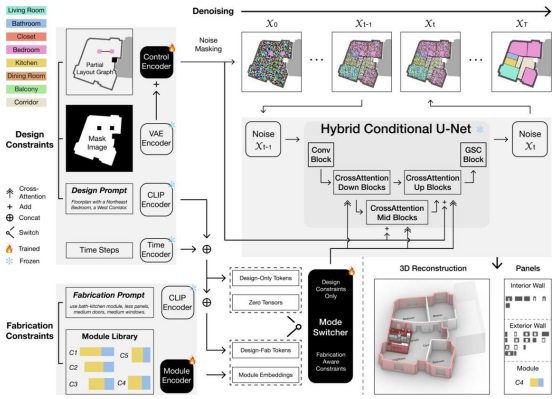
FabPLAN

Fabrication-aware generative design for hospital wet-core layouts

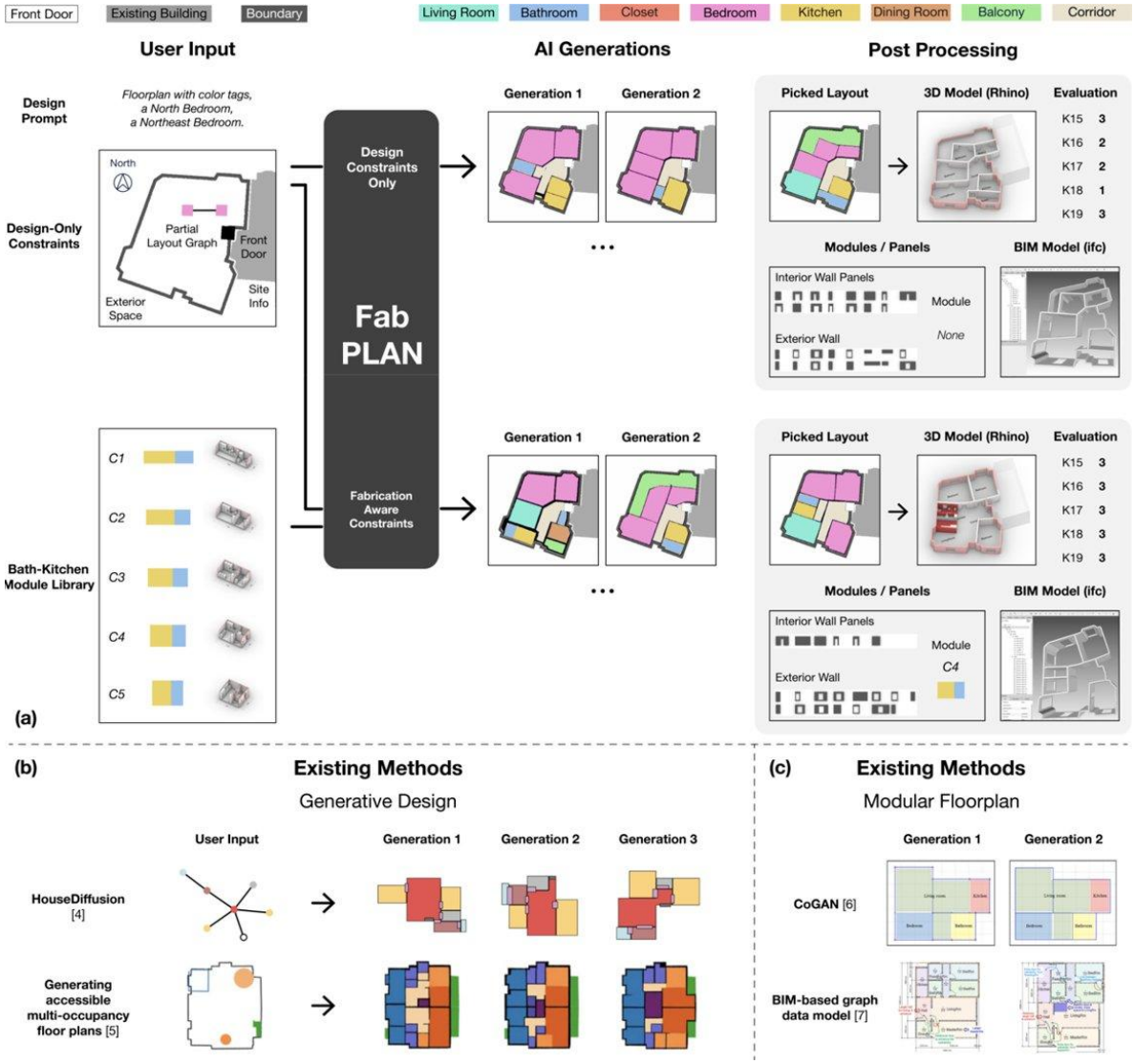
Under review

Prefabrication constraints become generative conditions.

FabPLAN targets the **bath–kitchen wet core**, where spatial quality and module fabrication must be resolved together. A **label-free data bootstrap** creates aligned design and fabrication examples; **two-stage training** supports **design-only** exploration and **fabrication-aware** generation.



Control and CLIP encoders feed a hybrid conditional U-Net.

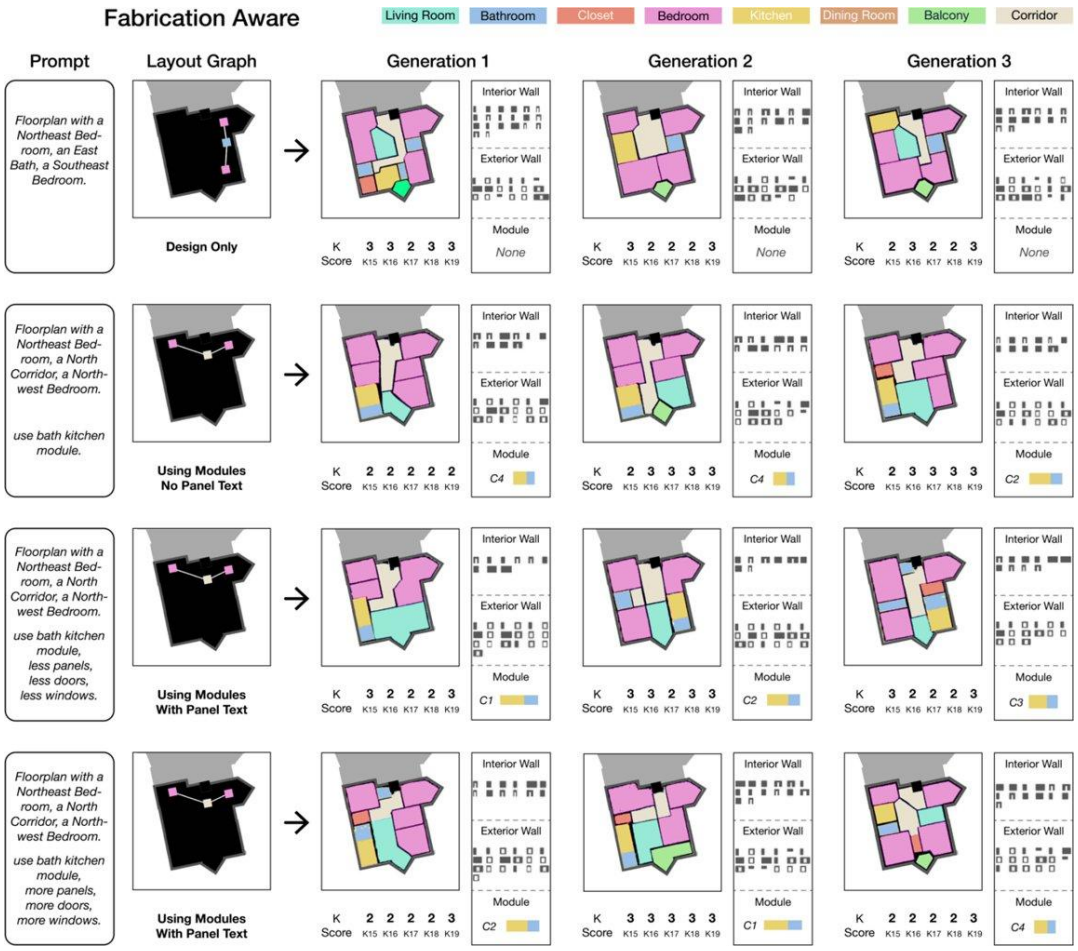


FabPLAN: fabrication-aware generation for hospital bath–kitchen wet cores.

PROJECT POSTER ↗

FabPLAN

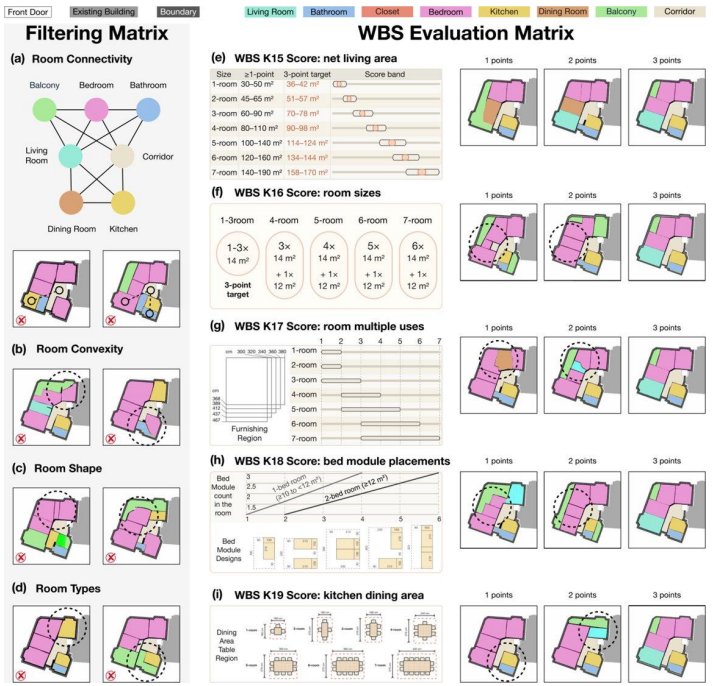
Iterative layout customization with fabrication evidence in the loop



Fabrication-aware generation under increasing module and prompt specification.

A generated plan is evaluated as a construction system.

A **WBS-inspired matrix** compares room geometry, module count, connectivity, and **fabrication feasibility**. Reconstruction tests whether wet cores can be paired with **modular elements** and assembled as coherent **3D systems**—making buildability visible before downstream detailing.



Geometry filtering and WBS proxy evaluation of decoded rooms.

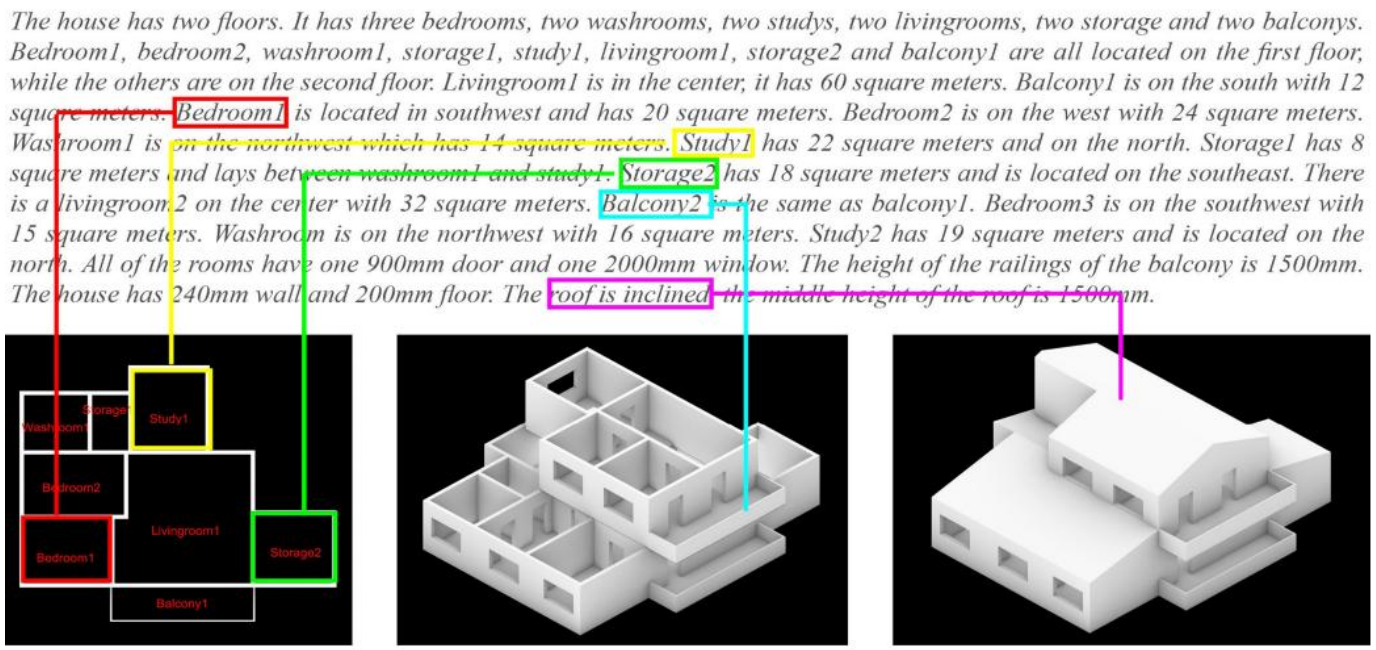
Text-to-Form

Language-driven 3D form generation through spatial graphs

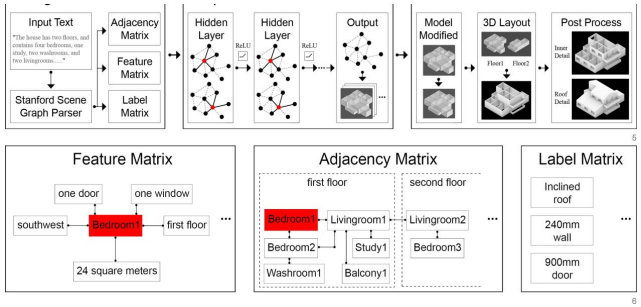
ACADIA 2020 · FIRST AUTHOR

A spatial sentence becomes an architectural massing.

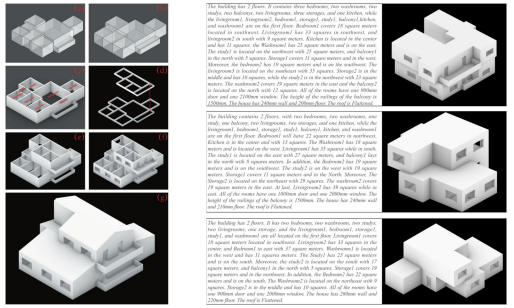
The Language-to-Form Generation Method parses room descriptions into a **scene graph**, encodes relationships with a **two-layer graph convolutional network**, and predicts 3D layout coordinates. Trained on **1,062 paired language–form samples**, the workflow connects semantic relations to editable **Grasshopper geometry**.



Highlighted room relations in language correspond to generated spatial arrangements.



Linguistic parsing and a GCN predict a structured 3D layout.



Post-processing turns graph coordinates into complete architectural models.

READ FULL PAPER ↗

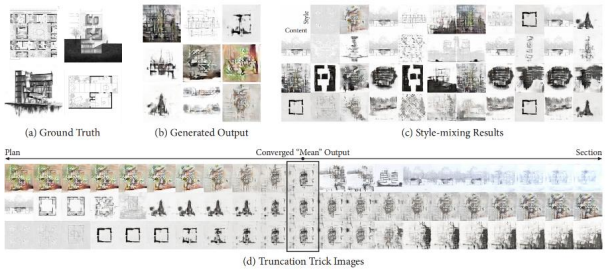
3D Model Generation

Architectural form from plan and section training

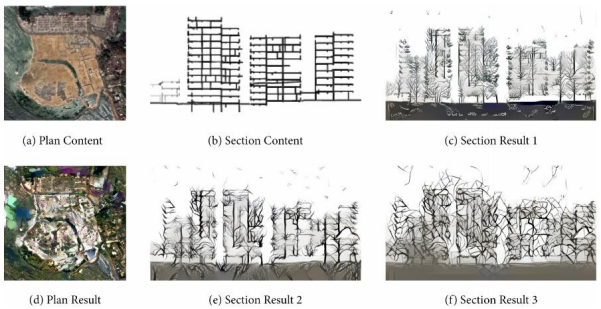
TECHNOLOGIES 2019 · FIRST AUTHOR

Train in 2D; synthesize architectural form in 3D.

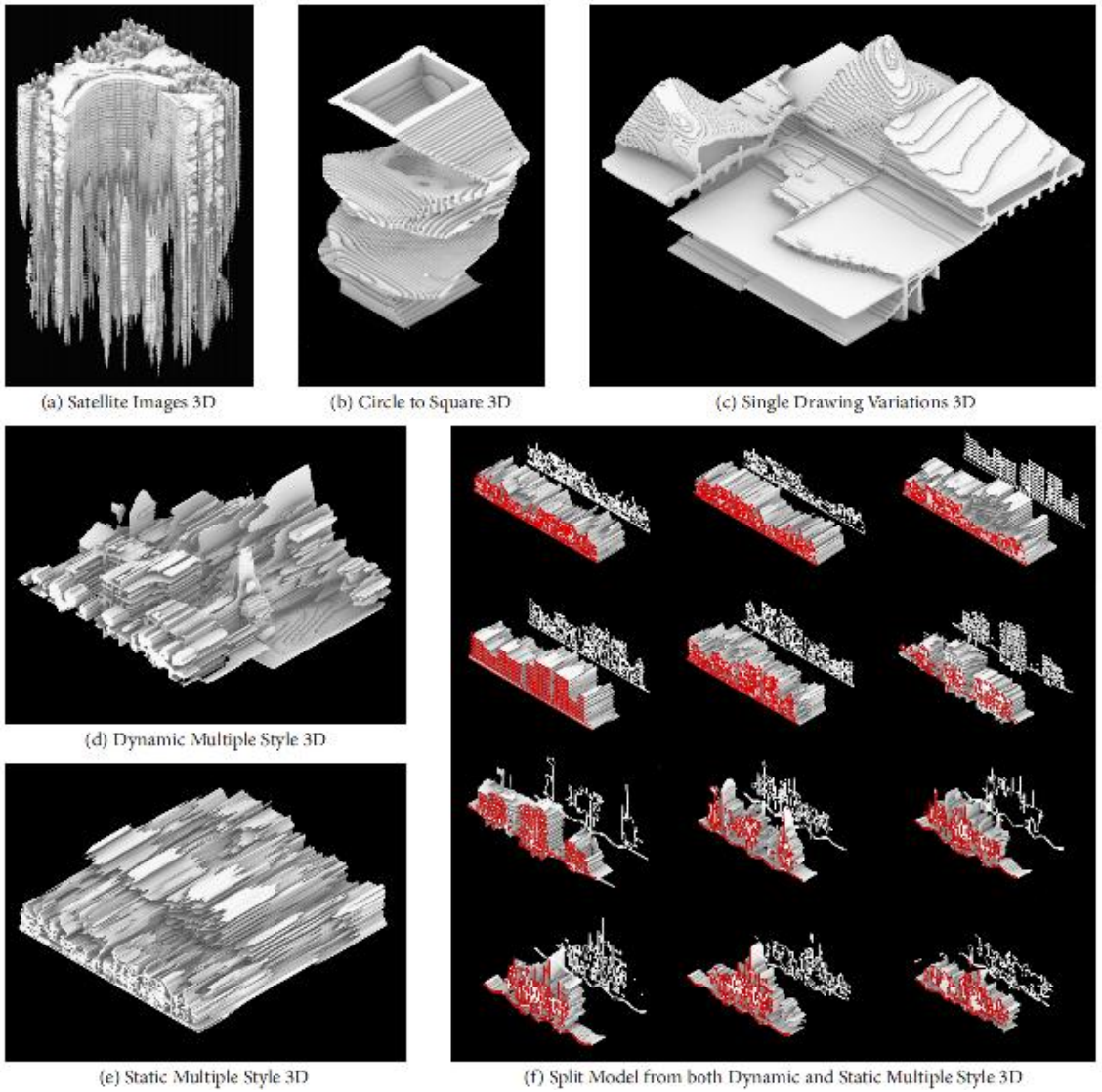
Plan and section drawings are serialized as image sequences and learned with a **generative adversarial model**. Sampling and interpolating the **latent space** creates transformation sequences that can be reconstructed into **voxel-based architectural forms**—an early bridge between representation learning and spatial design.



Plan and section drawings become aligned training sequences.



plan and section drawing training examples



Generated plan/section sequences are assembled into novel 3D architectural forms.

READ FULL PAPER ↗

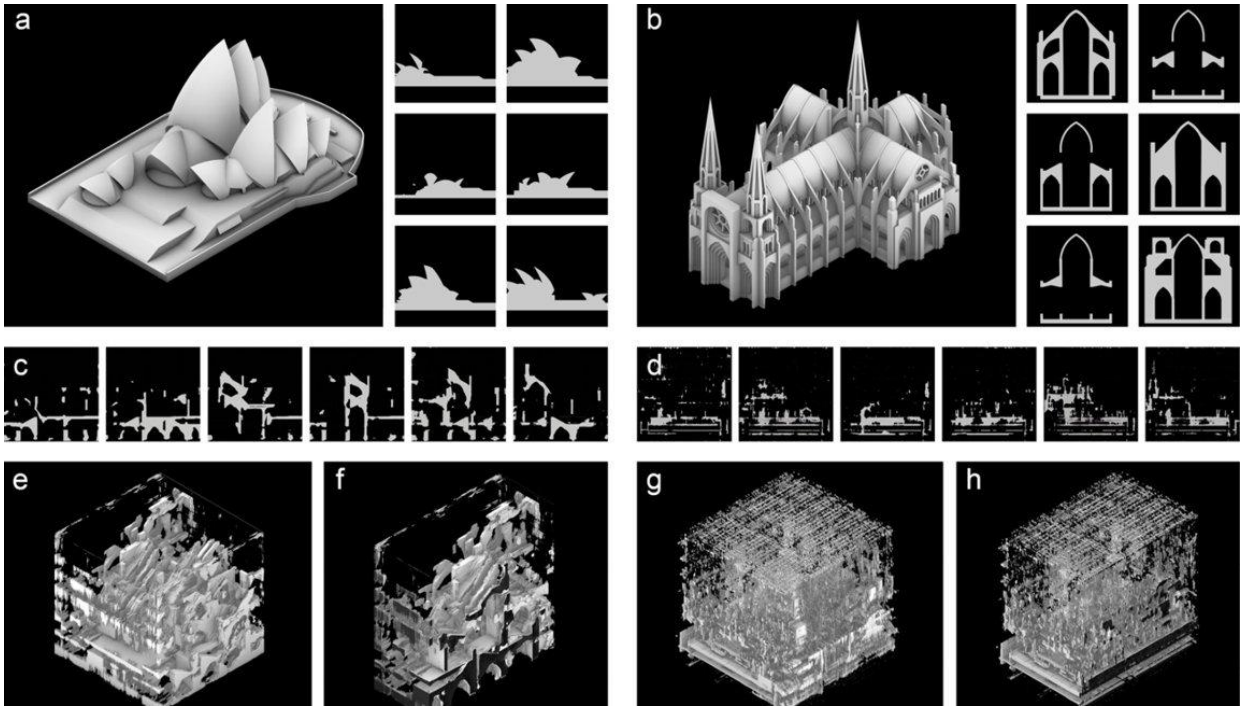
3D Architectural Form Style Transfer

Learning continuity across serial stacks and multiple views

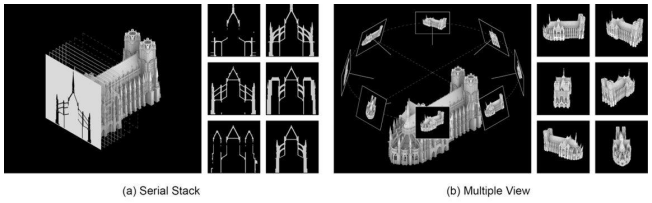
CAADRIA 2020 · FIRST AUTHOR

2D style transfer becomes a 3D reconstruction strategy.

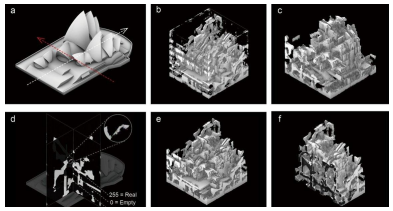
The study decomposes models into **serial stacks** or **calibrated multiview images**, applies **CycleGAN, Pix2Pix, and neural style transfer**, then rebuilds 3D form. **Continuity-aware preprocessing** reduces abrupt slice-to-slice changes and preserves more coherent spatial structure.



Style-transfer results via CycleGAN with continuity preprocessing.



Two deconstruction strategies: image-stack and multi-view to 3D.



Multi-axis style transfer through CycleGAN.

READ FULL PAPER ↗

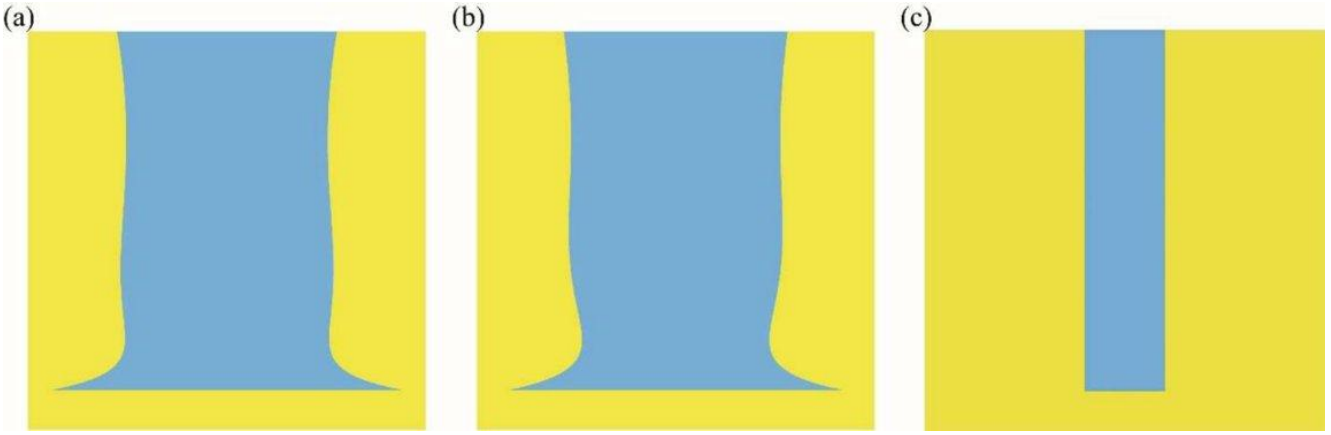
Composite Pillar Optimization

Machine learning for dry-adhesive microstructure design

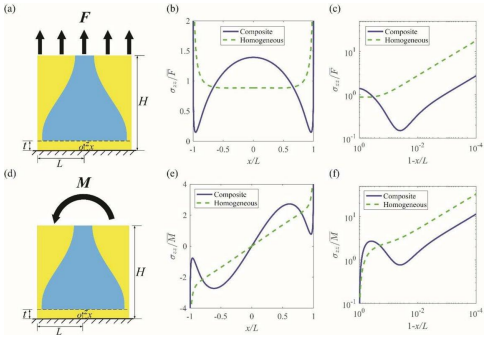
EXTREME MECHANICS LETTERS 2022 · CO-AUTHOR

A fast surrogate enables constrained mechanical search.

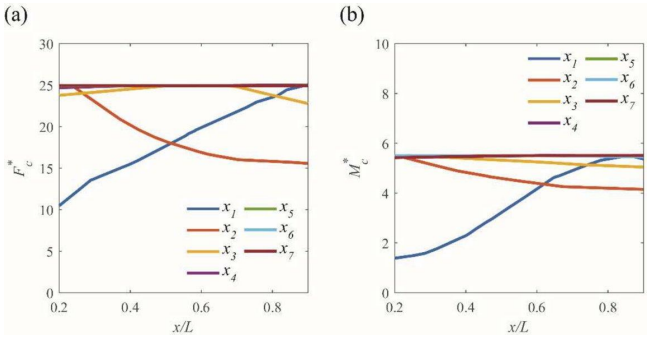
Neural networks trained on **finite-element simulations** predict adhesion with **average error below 1%**, enabling **constrained optimization** over composite cores. The reported best normal-loading design reaches **nearly 11×** the critical force of a homogeneous pillar and **about 1.7×** that of a simple wide rectangular composite.



Optimal composite-pillar designs under different performance criteria.



Rectangular composite pillar under normal loading; stiff core in blue.



Critical force and moment vs. each control-point position.

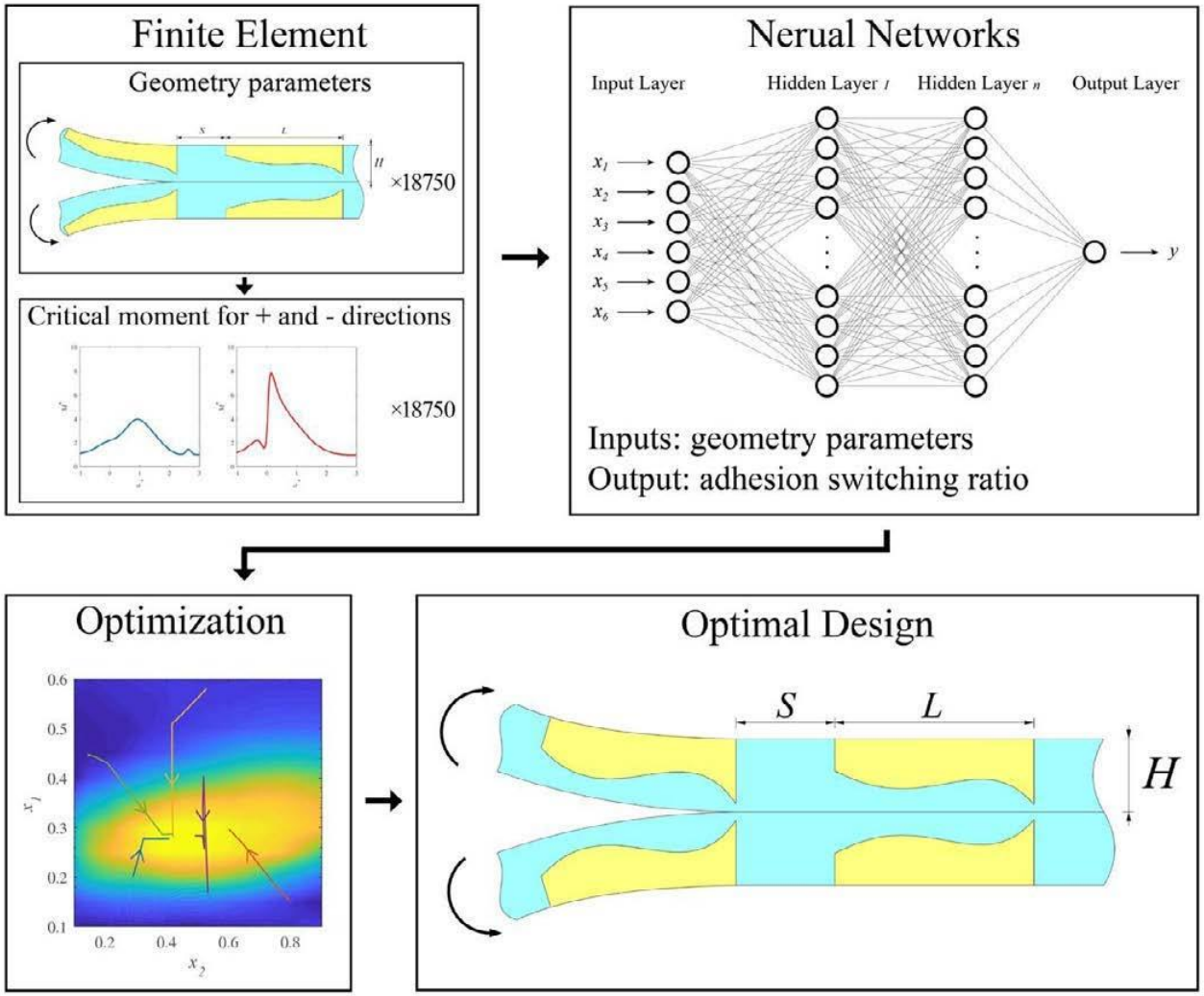
READ FULL PAPER ↗

Directional Adhesion

Asymmetric subsurface structures in double cantilever beams

Under review

Asymmetric heterogeneous insets beneath a homogeneous interface produce different adhesion capacities when a beam is loaded from opposite directions. Double cantilever beam tests compare L-shaped insets numerically and experimentally. A neural surrogate and constrained optimization search curvy inset designs. The highest reported adhesion switching ratio was 1.97. Shorter insets or thicker compliant layers reduce the ratio.



Simulation, surrogate model, and optimized inset geometry.

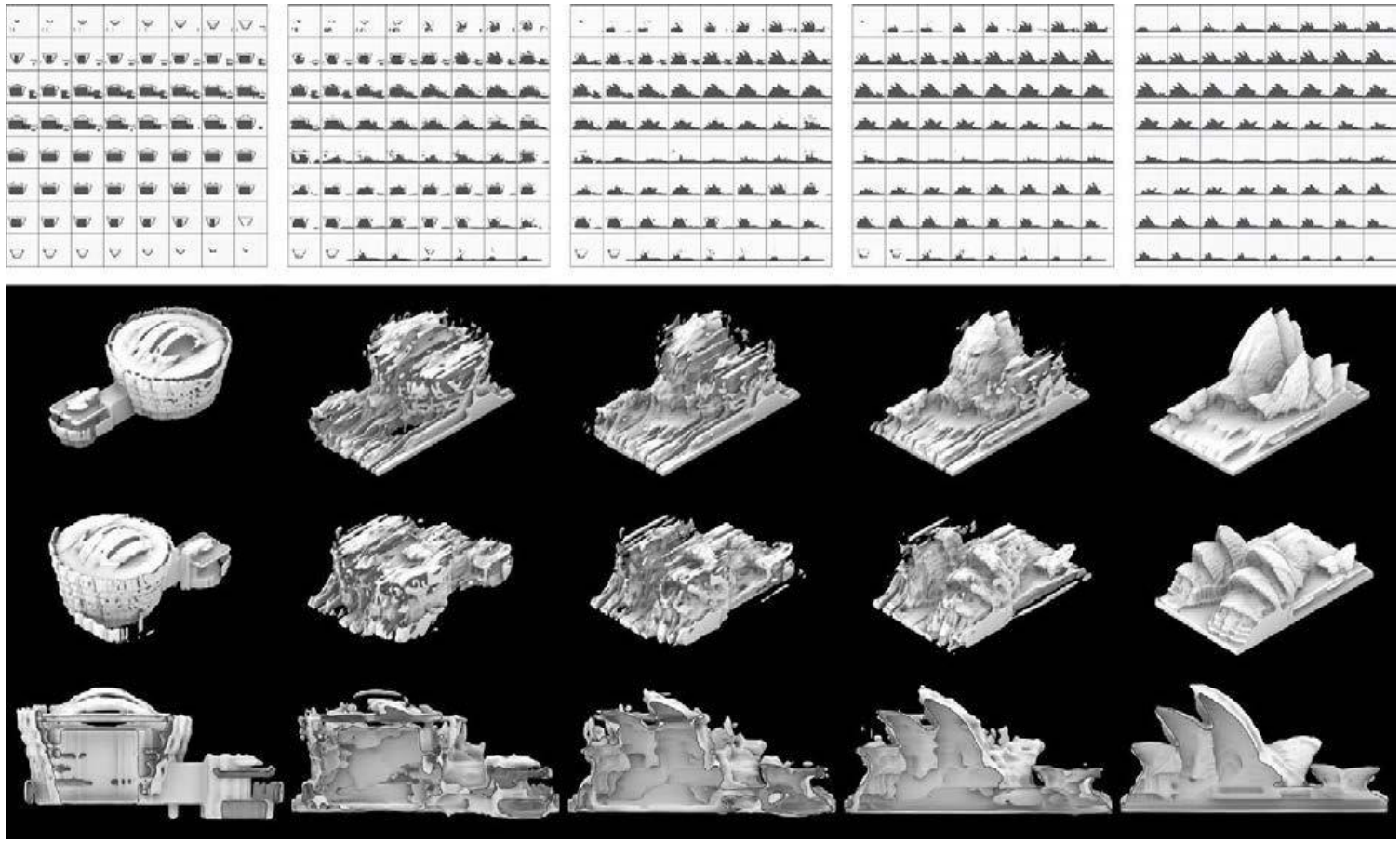
Numerical testing, experiments, and constrained design search

2D–3D Form Finding

High-resolution generation through slice encoding and refinement

DigitalFUTURES CDRF 2020

A 2D–3D encoding workflow splits architectural models into ordered section images for StyleGAN training. A 64×64 main-network output is enlarged with Waifu2X and refined using paired high-resolution data in Pix2Pix. The study also arranges 64 sections into an 8×8 image to encode a model's spatial sequence. Generated intermediate forms blend morphological features across source buildings.



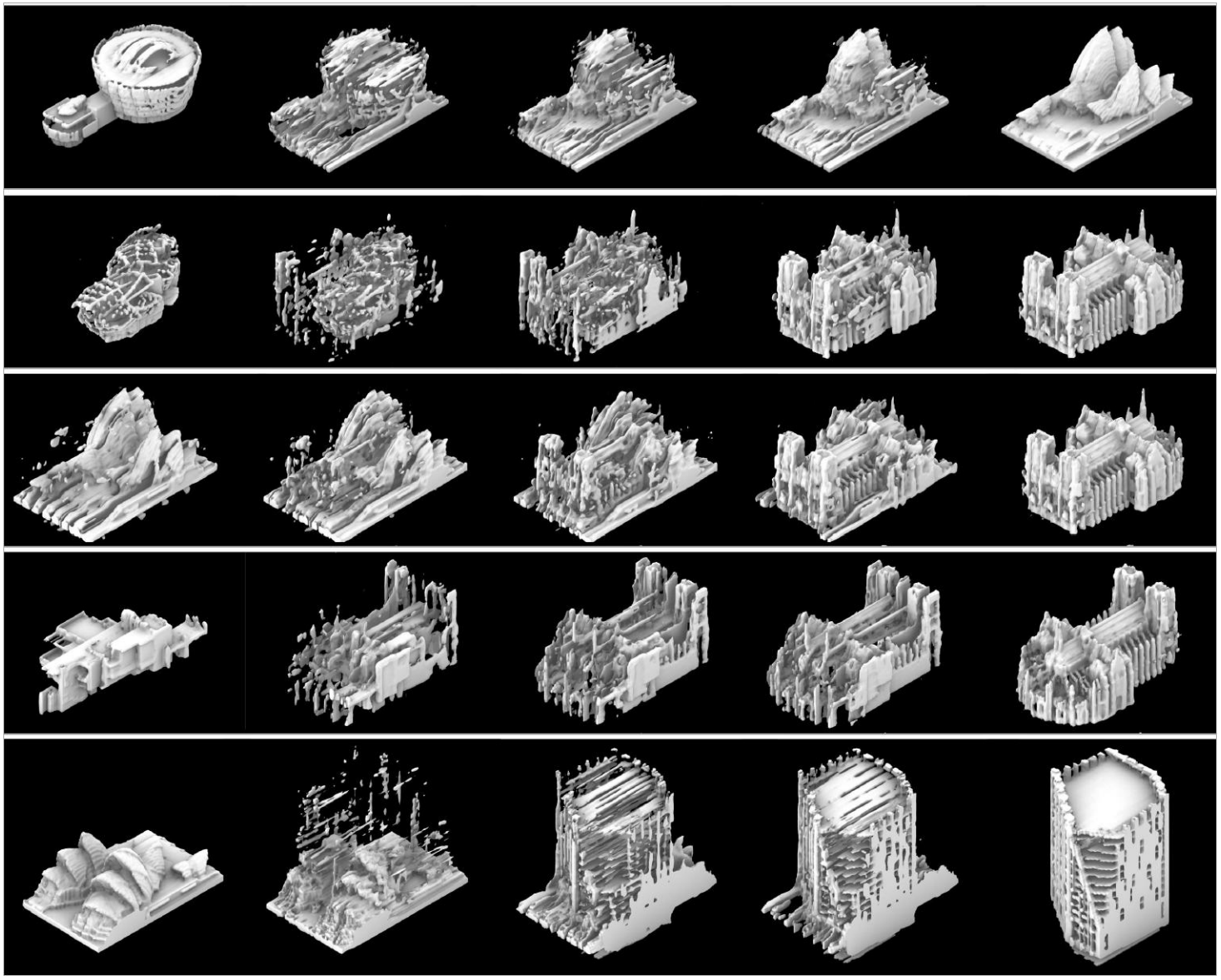
Intermediate architectural forms reconstructed from image sections.

Section encoding, resolution enhancement, and form reconstruction

2D–3D Form Finding · Intermediate Forms

Generated transitions between distinct architectural forms

DigitalFUTURES CDRF 2020

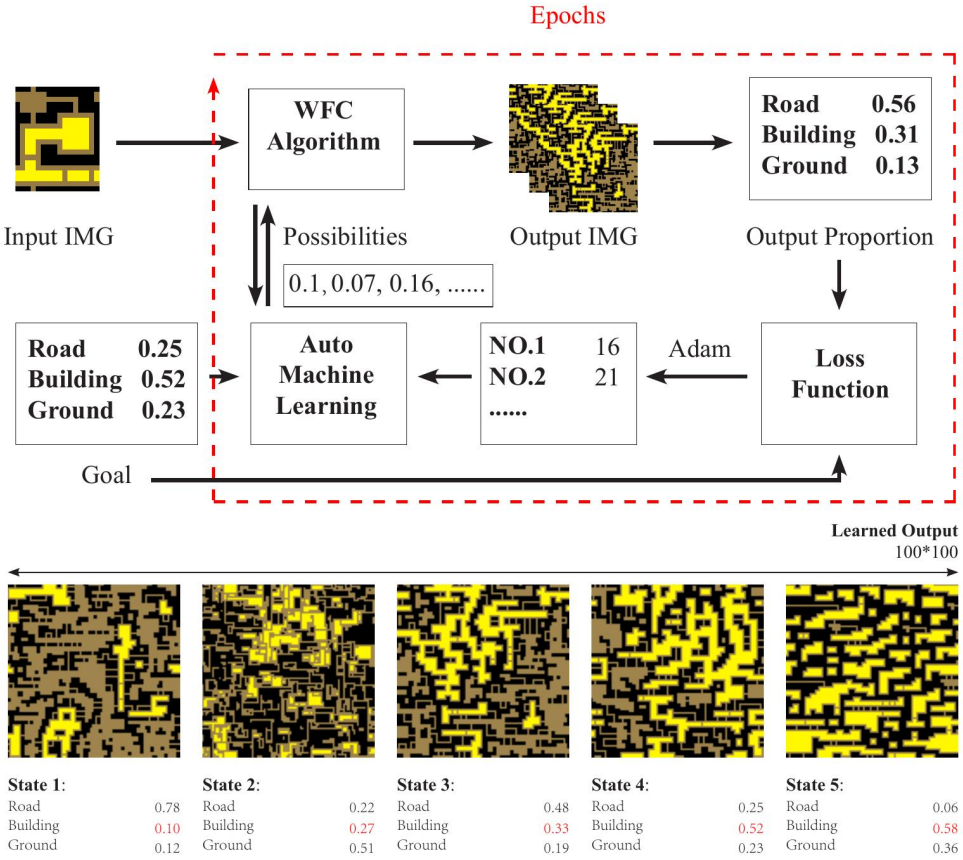


Rows of intermediate 3D forms show the gradual blending of geometric features.

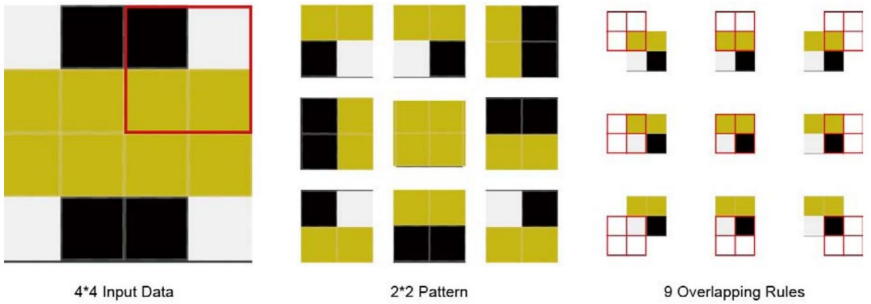
Wave Function Collapse

Pattern analysis and goal-directed generation

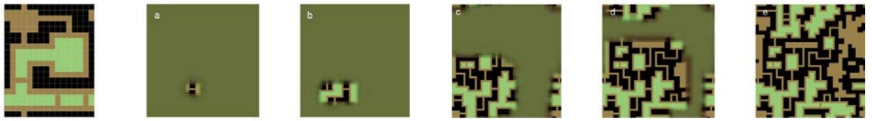
CAADRIA PGSC 2020



WFC output is measured against target proportions. An AutoML loop updates pattern probabilities across epochs.



Overlapping 2 × 2 patterns define local adjacency rules.

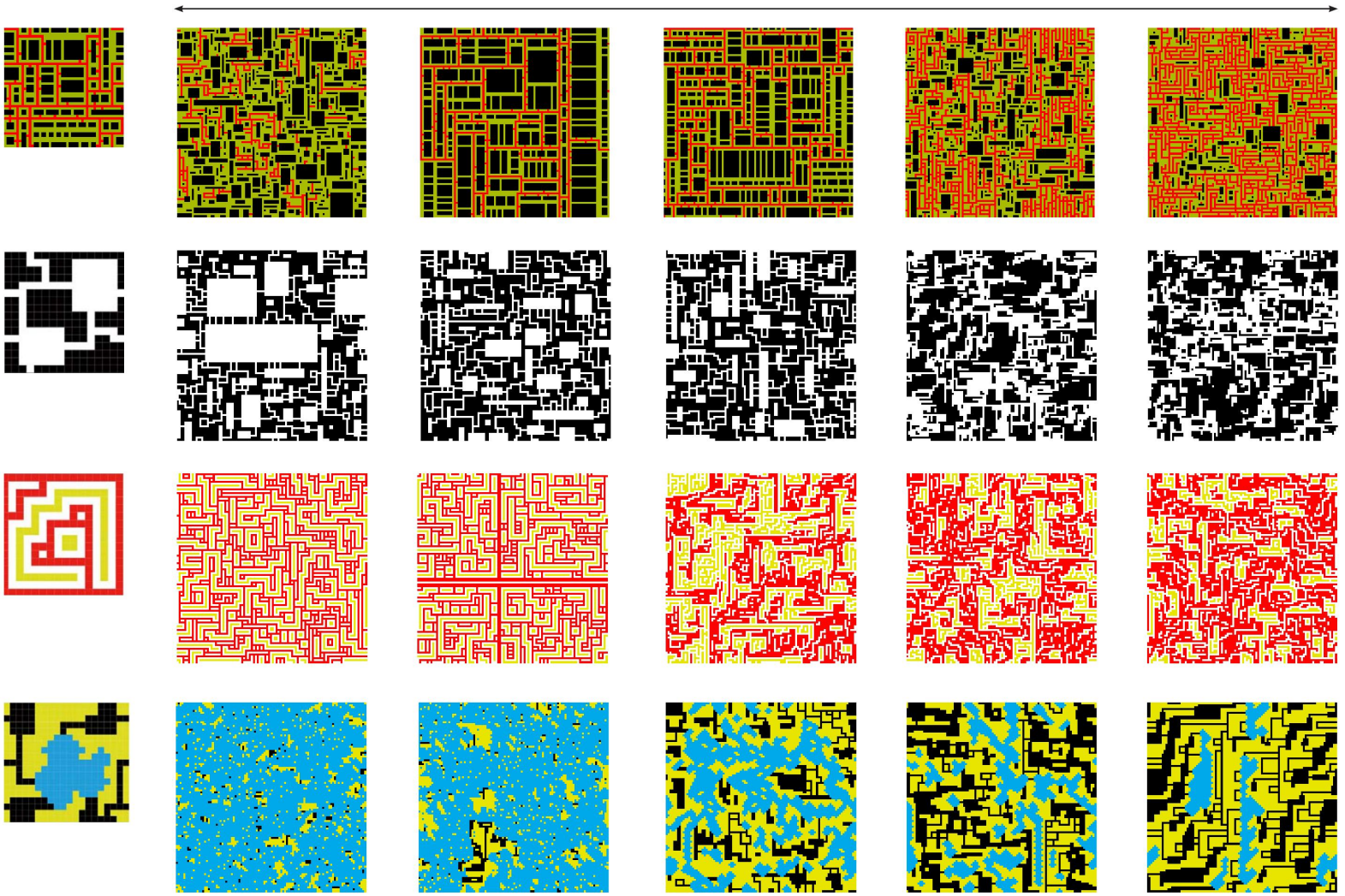


Each observation collapses a low-entropy region, then propagates the resulting constraints.

Wave Function Collapse · Results

Generated variations from four input patterns

CAADRIA PGSC 2020



Each row starts with an input pattern and shows five generated variations that retain its local rules.

DESIGNS

FUTURE AIRPORT

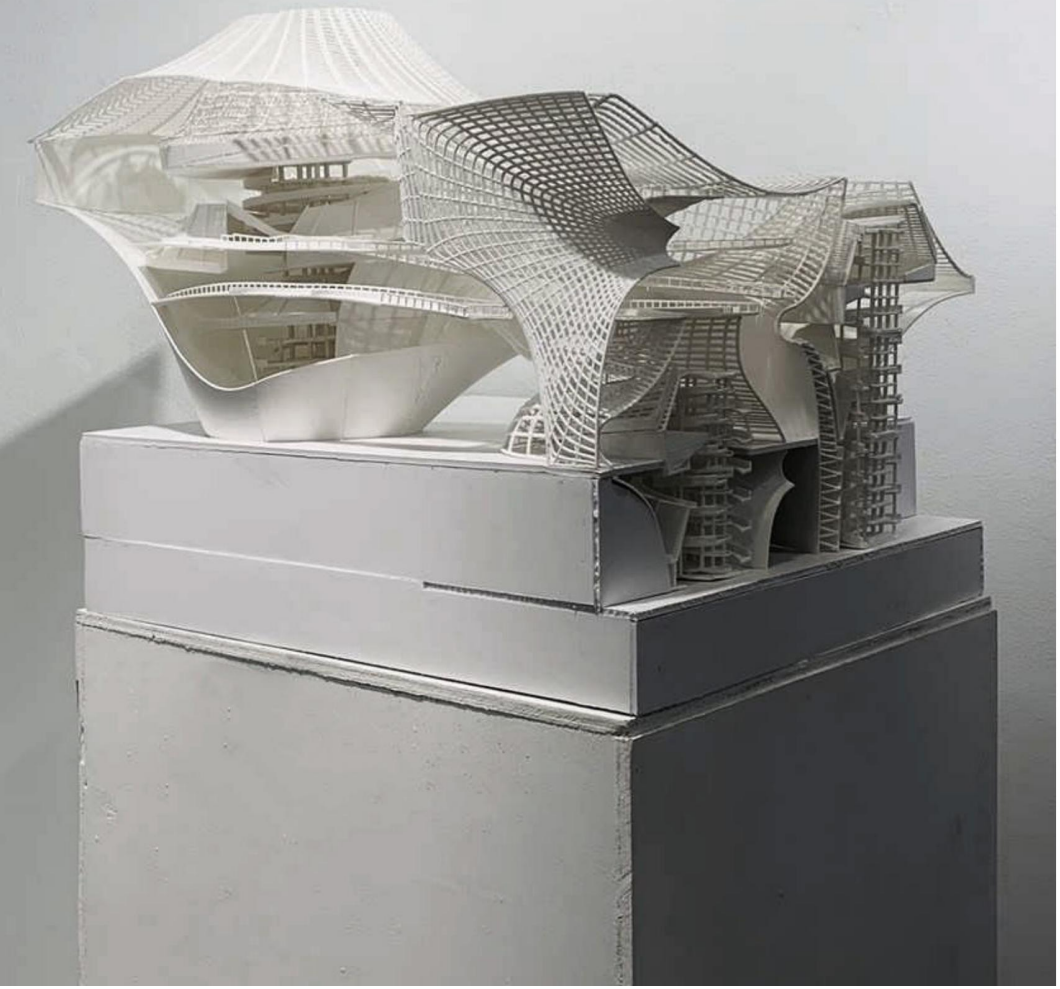
Academic Project

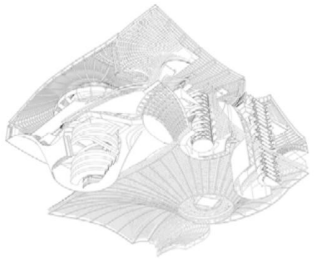
Instructed by Ali Rahim

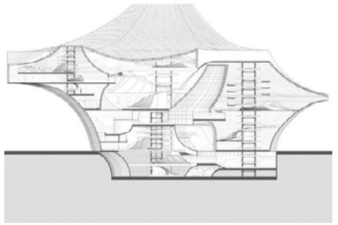
Fall, 2018

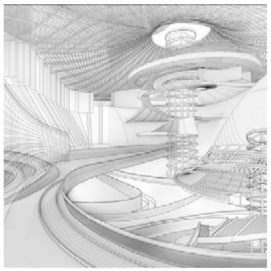
Team Work

The professor hopes to integrate logistics system into the airport space and influence the final structure with the Yoyogi National Gymnasium as a reference. I don't think this design logic make enough sense, but hey we've made a fantastic model.

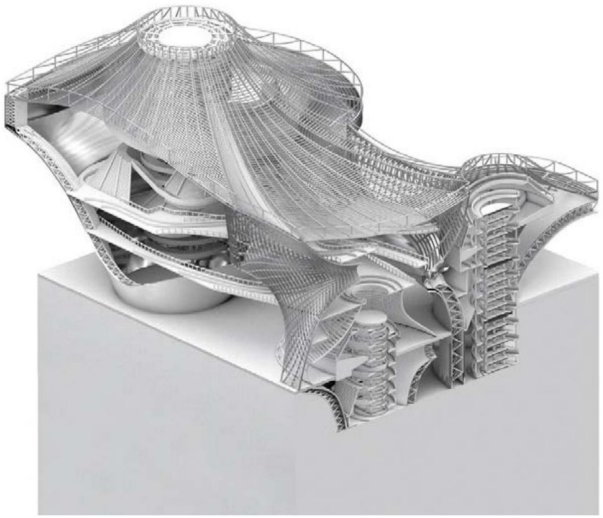


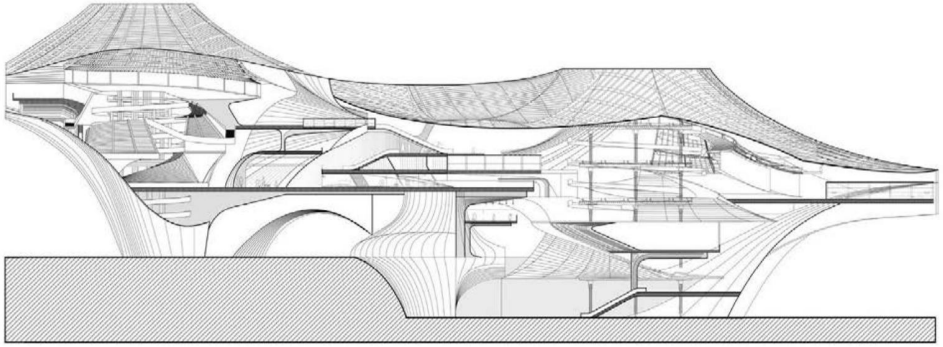












LIGHT TRACING

Academic Project

Instructed by Sirun Hu

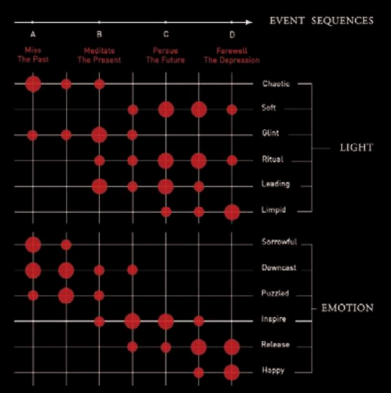
Fall, 2016

Individual Work

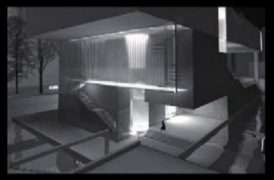
There are possibilities for architecture to transfer their emotion, which is within their spaces and structures, to these vistors who also hold their own emotions. No matter what emotion these people might have before, it is the place that offers a chance to make the emotional interaction. During our journey within these architecture, we can get in touch with them through these forms of light.





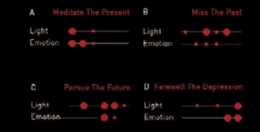


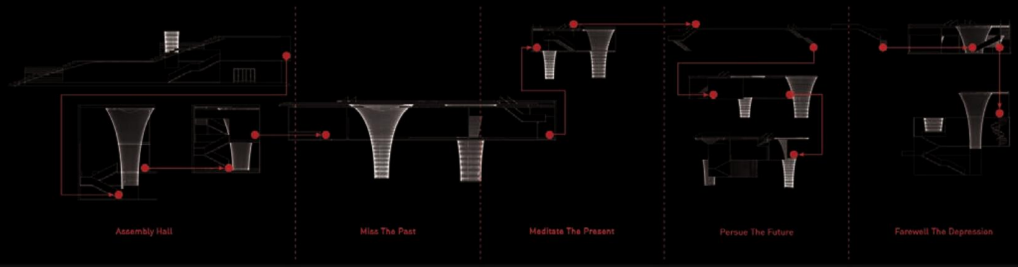


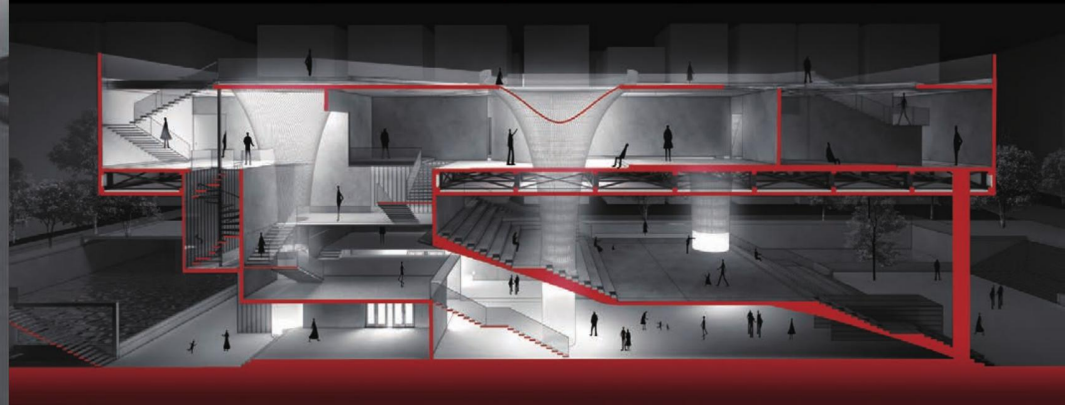


EVENT WITH LIGHT AND EMOTION

In the main four event sequence, there are several kinds of light they require, also different level for the degree of the light are measured. During the journey through these sequences, they will raise their spirit from negative to positive.



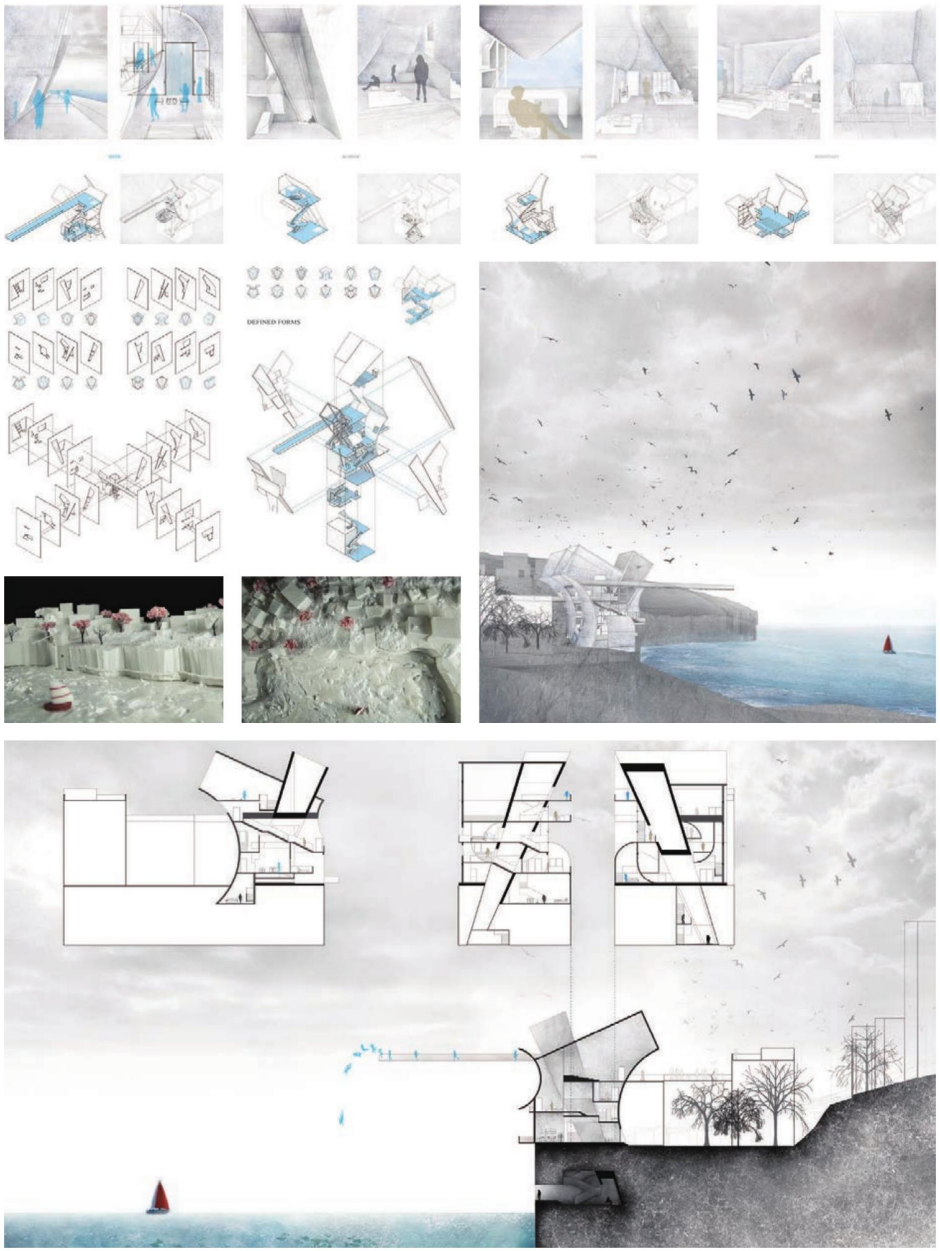




SCHIZOBULIA HOUSE

Site : Fu Jian, China
Type : Academic Project
Advisor : Hanqing Zhang
Summer , 2016
Individual Work

Modern society has been searching for the existence and possible cure of Schizobulia - multiple personalities. However, as the biggest unusual feature compared to normal people, split and co-existing personalities are the most unique characteristics of them. Instead of pulling them back to the normal state of mind, along with pain and suffer, it is better to adapt and satisfy their personalities to provide a living environment for their personalities to coexist.





DIFFUSION STRUCTURE

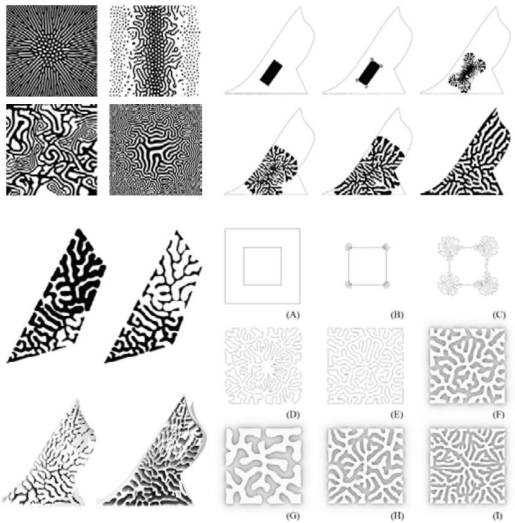
Seminar Project

Instructed by Akbarzadeh Masoud

Spring, 2019

Team Work

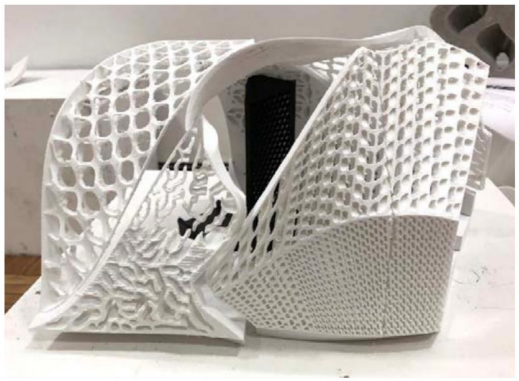
Diffusion reaction is an interesting generative morphological algorithm. For the application of this algorithm in the design and masonry of building walls, the specific force condition should be considered. According to the iteration degree of the generation algorithm and the adjustment of local thickness, we optimized the final wall generation result, controlled its three-dimensional shape in the rule of surface, so as to finally use the robotic arm to cut out the specific design shape and filled concrete to finalize the design.













FASHION IN ARCH

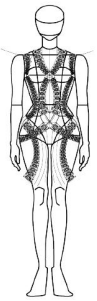
Seminar Project

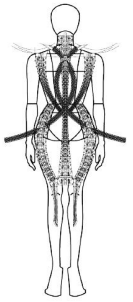
Instructed by Danielle Willems

Spring, 2019

Team Work

We try to explore the way of inspiring fashion clothing design based on architectural design, and made the clothing with skeleton and woven network by means of parametric design. In turn, we also integrate the materials and organizational characteristics of clothing production into the innovation of architectural spatial form and structure.

















Architectural intent becomes a controllable system.

My research develops generative models that accept incomplete ideas, preserve user agency, and expose fabrication consequences. The projects in this portfolio connect language, geometry, machine learning, and material systems to architectural design.

Hang Zhang · Digital Building Technologies · ETH Zurich

hangzhang@ethz.ch

github.com/HangZhangZ ↗

2026